

I . Answer:

1	2	3	4	5	6	7	8
B	D	D	C	A	C	A	B

II . Answer:

1、 x=2.333 y=2.000 z=1.000	2、 The number is: one two three four
3、 x=10 y=7 x=7 y=10	4、 1 1 1 2 1 3
5、 2 2 4 4 5	6、 BDBB
7、 2 6 10	8、 4 2 2 19 14 15 11 15 -4 34 31 3

III. Answer:

- (1) iomani
- (2) std
- (3) DoTower
- (4) circleCount>0
- (5) circleCount-1
- (6) endPag
- (7) auxPag
- (8) circleCount-1
- (9) beginPag
- (10) endPag

1. Example:

```
#include <iostream>

using namespace std;

long fact(int n)
{
    if (n < 2)  return 1;

    long f = 1;

    for (int i=2; i<=n; i++)

        f=f*i;

    return f;
}

long comb(int n, int k)
{
    if (n<0 || k<0 || k>n) return 0;

    return fact(n)/(fact(k)*fact(n-k));
}

int main()
{
    int n,k;

    cin >>n >>k;

    cout << comb(n,k)<<endl;

    return 0;
}
```

2. Example:

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string str, sstr;
    int cases,i;
    cin >> cases;
    for (i=0; i<cases; i++)
    {
        cin >> str >> sstr;
        int count = 0;
        int loc = 0;
        while (loc < str.length())
        {
            loc = str.find(sstr, loc);
            if ( loc == string::npos)
                break;
            else
            {
                count++;
                loc++;
            }
        }
        cout<<count<<endl;
    }
    return 0;
}
```

3. Example:

```
#include <iostream>
using namespace std;
int main()
{
    int cases,i,j,k,l,a,b;
    int arr[20];
    cin >> cases;
    for (i=0; i<cases; i++)
    {
        cin >> a;
        j = 0;
        while (a > 1)
        {
            if (a%2 == 0)
                arr[j] = 0;
            else
                arr[j] = 1;
            a=a/2;
            j++;
        }
        if (a==1)
            arr[j] = 1;
        else
            j--;
        k = 0;
        b = 0;
        for(l=j; l>=0; l--)
        {
            b = b + (arr[l] << k);
            k++;
        }
        cout << b << endl;
    }
    //system("pause");
    return 0;
}
```